1. Define conceptual schema.
2. What is data independence?
3. Define an entity set.
4. What are attributes in ER model?
5. Define relationship set.
6. What is logical database design?
7. What is the purpose of foreign key constraints?
8. Write any one difference between UNION and INTERSECT.
9. What is a view in the relational model?
10. Write any one difference between tuple relational calculus and domain relational calculus.
11. Mention one use of the UNION operator in SQL.
12. Define a functional dependency.
13. What is a nested query?
14. Define BCNF.
15. What is cascading rollback?
16. Define clustering index.
17. Mention one difference between B Trees and B+ Trees.
18. Mention any one aggregate function in SQL.
19. Define 3NF
20. What is a transaction?
21. Define recoverability in schedules.
22. What is indexing?
23. What is hashing in indexing?
24. What is “order” in B+ trees?

1. Explain data abstraction levels with examples.
2. Discuss the architecture of a DBMS with a neat diagram.
3. Explain the historical perspective and evolution of DBMS.
4. Design an ER diagram for a university enrollment system given specifications

5. Write notes on file systems vs DBMS.
6. Explain the structure of DBMS and its main components.
7. What are integrity constraints? Explain types with examples.
8. Write relational algebra queries: i) List all employees earning more than average salary. ii) Find all departments that have no employees. Iii) Find all employees in department 8 who earn more than 3000. State assumptions made
9. What are views in SQL? Explain with examples.
10. Define domain relational calculus and give a query example.
11. Differentiate between domain relational calculus and tuple relational calculus.
12. Explain schema modification in SQL.
13. Explain how integrity constraints are enforced in SQL.
14. Write SQL queries for: i) Find employees working under manager 'John'. ii) List all employees with NULL commission.
15. Write SQL statements to: i) Find departments with more than 5 employees. ii) List employees hired before 2020. Use the following schema

    EMPLOYEE(EmpID, EmpName, HireDate, DeptID)

    DEPARTMENT(DeptID, DeptName)
16. Write SQL queries for: i) Display employees who earn the highest salary. ii) Find employees who do not have managers.

    Employee(emp_id, name, salary, manager_id)
17. Given

StudentID	**StudentName**	**Courses**	**Instructor**	**InstructorPhone**
1	Yadav	DBMS, AI	Dr. Smith, Dr. Roger	555111, 555222
2	Hari	AI	Dr. Roger	555222
3	Neha	DBMS, Networks	Dr. Smith, Dr. Ken	555111, 555333

    a. Flatten multi-valued fields and Identify multivalued dependencies → go up to 3NF
18. Define multivalued dependencies and explain 4NF with an example.
19. A relation STUDENT_COURSE has the following attributes: (5M)

    STUDENT_COURSE(StudentID, StudentName, CourseID, CourseName, InstructorName, InstructorOffice)

    Assume the following functional dependencies hold:

    StudentID → StudentName

    CourseID → CourseName, InstructorName

InstructorName → InstructorOffice

i. Identify all the partial and transitive dependencies.
ii. Normalize the relation up to **Third Normal Form (3NF)**, clearly showing each step of decomposition.

20. What are complex integrity constraints? Explain with examples.
21. Explain fifth normal form with suitable example.
22. Explain triggers and their components with an example.
23. Discuss anomalies that can occur in a non-normalized table.
24. Explain the states of a transaction.
25. What is serializability? Explain with precedence graph.
26. What is the equivalent serial schedule for the following transactions?
27. Discuss two-phase commit protocol and its types.
28. Write a short note on deadlock prevention in DBMS.
29. Differentiate between undo and redo in recovery.
30. Describe different file organization techniques.
31. Explain tree-based indexing with reference to B+ Trees.
32. Write a note on secondary indexes and their significance.
33. What is a lock-based protocol? Explain two-phase locking protocol.
34. Explain the process of automatic acquisition of locks in SQL.
35. What are the different concurrency control protocols. Explain the pitfalls of lock based protocols.
36. Discuss how external data storage affects indexing.
37. Compare clustered and non-clustered indexes.
38. What are the various deadlock prevention strategies? Explain.
39. Discuss recovery techniques when multiple transactions are concurrent.
40. Write short notes on: i) Primary Index ii) Secondary Index
41. Explain the structure and operations of B+ Trees with a diagram.
42. Compare heap, sorted, and hashed file organizations.
43. Show the B+ tree that would result from successively deleting the data entries with keys 20, 25, 30,32 and 38 from the original tree.

